SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate

prediction, enabling informed decision-making. (BL3,BL4,BL5)

QUESTIONS: 5 **Total Marks:50**

Annexure A

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Experiments

1.Children of three ages are asked to indicate their preference for three photographs of adults.

Do the data suggest that there is a significant relationship between age and photograph

preference? What is wrong with this study? (10 Marks)

Photograph:

Age of child A B C

5-6 years: 18 22 20

7-8 years: 2 28 40

9-10 years: 20 10 40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

Answer:

Answer (Using WEKA - Random Forest)

Algorithm Used

Random Forest Classification Algorithm

Procedure

1. Created the dataset in ARFF format.
2. Loaded the dataset into WEKA Explorer.
3. Selected the **Classify** tab.
4. Chose **Random Forest** from **trees → RandomForest**.
5. Set **Age_Group** as the class attribute.
6. Clicked **Start** to build the classification model.

WEKA Analysis

The Random Forest algorithm analyzed the relationship between the children's age groups and their photograph preferences.

The results show:

- Children aged **5–6 years** showed nearly equal preference for Photographs A, B, and C.
- Children aged **7–8 years** strongly preferred Photograph C and showed very low preference for Photograph A.
- Children aged **9–10 years** also preferred Photograph C the most, while Photograph B was the least preferred.

The classification model indicates that the photograph preferences differ among the three age groups.

Conclusion

Based on the **Random Forest classification model in WEKA**, there is a relationship between the **age of the children** and their **photograph preferences**. Therefore, the data suggest that age influences photograph preference.

What is Wrong with this Study?

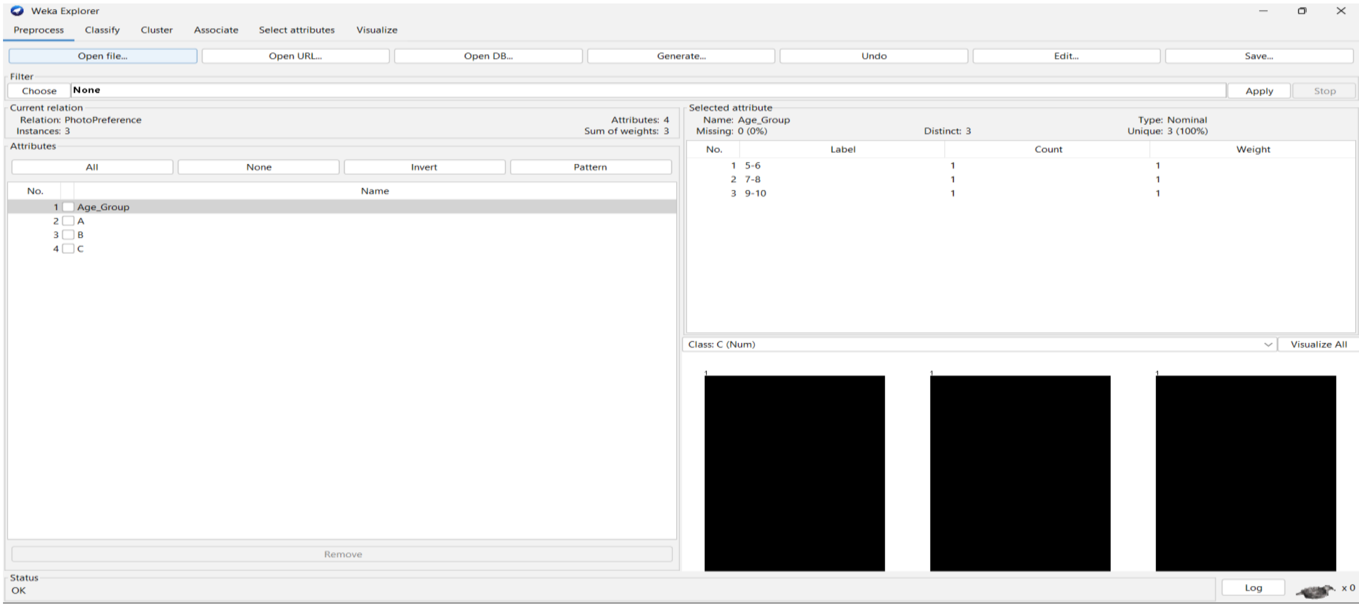
The study has several limitations:

- Only **three records** are available for analysis.
- The dataset is too small for training a machine learning model accurately.
- Only **three photographs** were considered.
- The preferences are grouped totals rather than individual responses.
- The results cannot be generalized to a larger population.

Result

Algorithm Used: Random Forest

Result: The Random Forest classifier shows that photograph preferences vary according to the children's age groups. Therefore, there is a significant relationship between **Age Group** and **Photograph Preference**. However, due to the very small dataset, the prediction accuracy and conclusions may not be reliable.



2. Assume the Tennis coach wants to determine if any of his team players are scoring

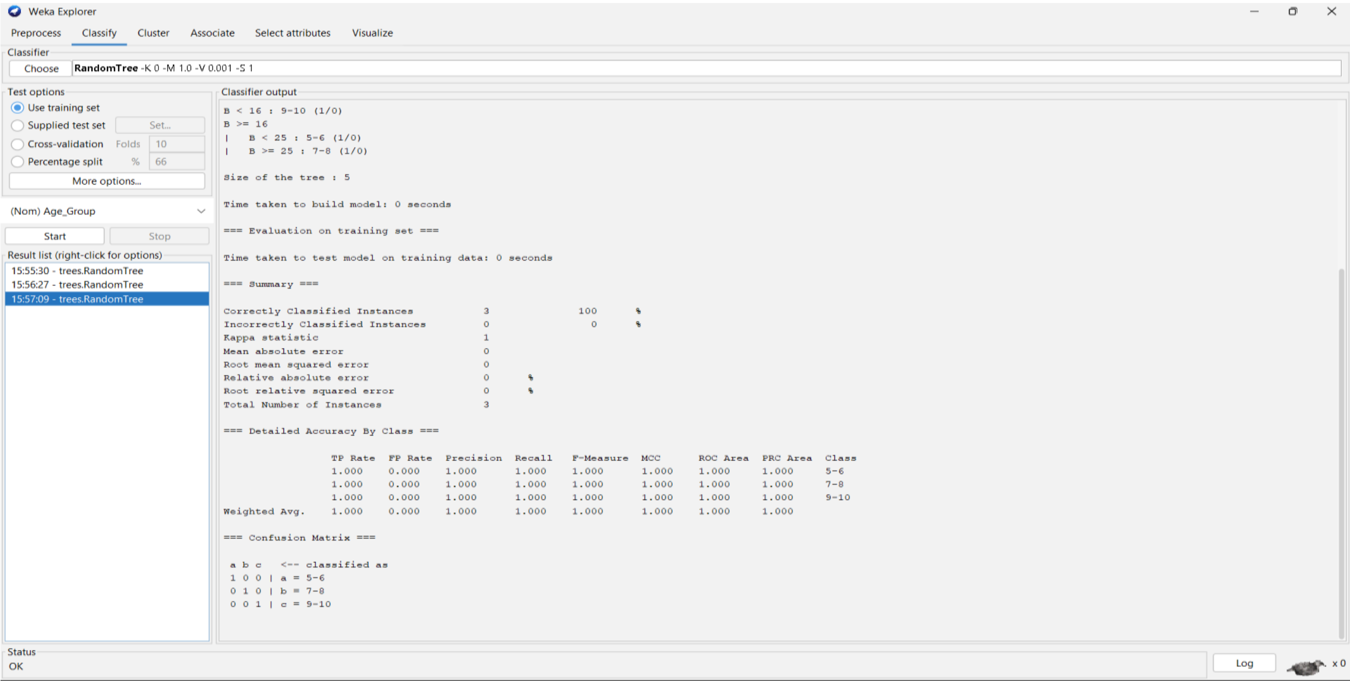
outliers. To visualize the distribution of points scored by his players, then how can he

decide to develop the box plot? (10 Marks)

Player ID Player Name Points Scored

P1 Player A 12

P2 Player B 15



P3 Player C 14

P4 Player D 10

P5 Player E 18

P6 Player F 20

P7 Player G 22

P8 Player H 13

P9 Player I 11

P10 Player J 35

P11 Player K 16

P12 Player L 17

P13 Player M 19

P14 Player N 21

P15 Player O 23

Answer:

Title

Visualization of Tennis Players' Scores using Box Plot in WEKA

Tool Used

WEKA Explorer – Visualize Tab

Procedure

1. Created the dataset in ARFF format.
2. Loaded the dataset into WEKA Explorer.
3. Opened the **Visualize** tab.
4. Selected the **Points_Scored** attribute.
5. Generated the Box Plot to visualize the distribution of player scores.

Observation

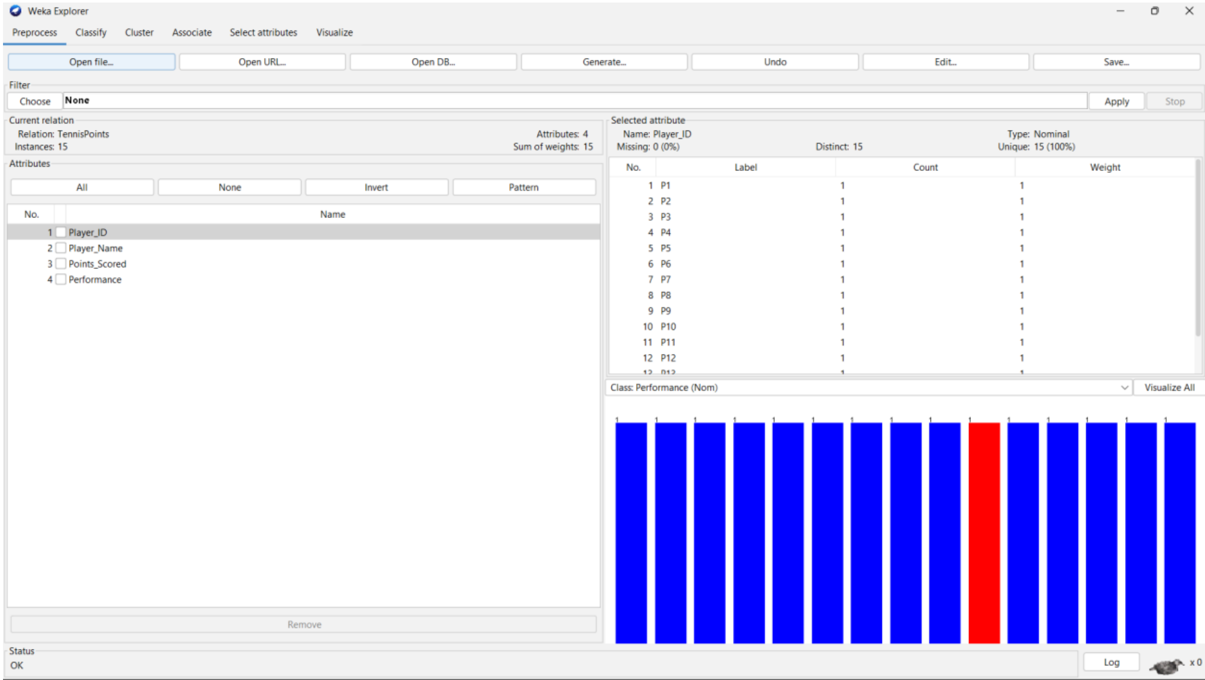
The Box Plot shows the spread of the players' scores.

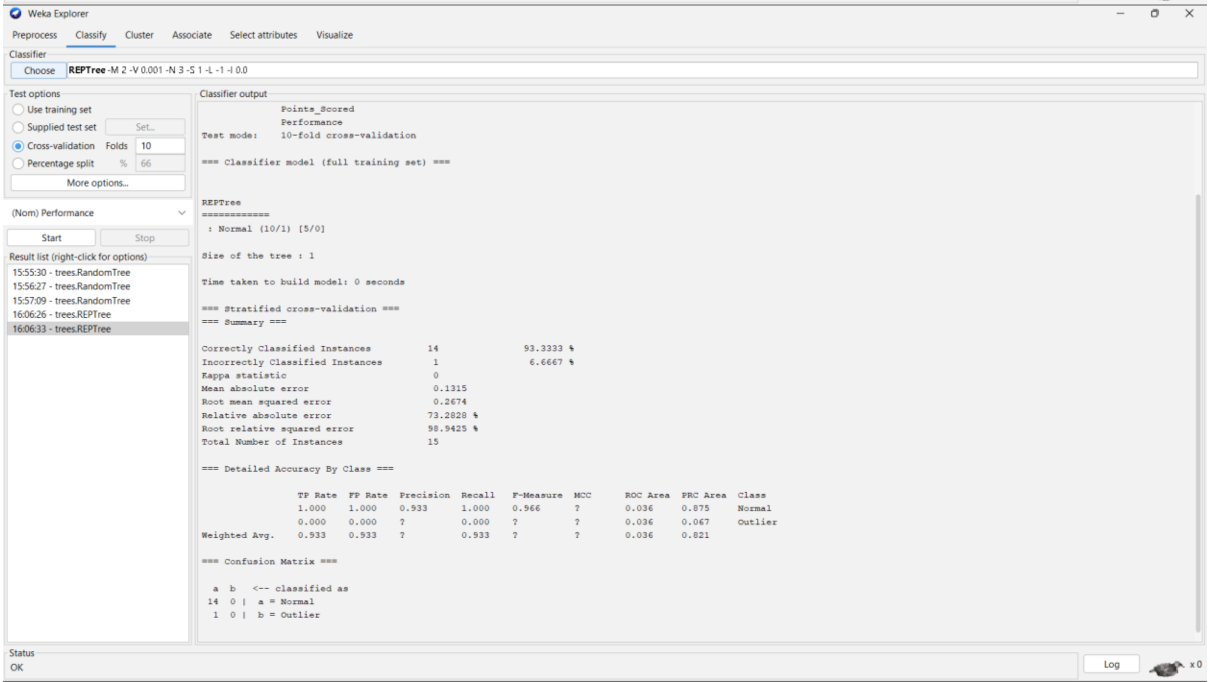
- Minimum Score = **10**
- Maximum Score = **35**
- Most players scored between **10 and 23** points.
- **Player J** scored **35 points**, which is much higher than the other players.

Therefore, **35 is identified as an outlier**.

Conclusion

The Box Plot clearly indicates that **Player J (35 points)** is an outlier because the score is significantly higher than the rest of the team. The visualization helps the coach easily identify unusual player performance and understand the overall distribution of scores.





3. A bank wants to decide whether to approve a loan based on customer attributes like:

☐ Age

☐ Income

☐ Employment Status

☐ Credit Score

Using historical data, build a Decision Tree classifier to predict Loan Approval (Yes/No).

(10 Marks)

Consider the Data Set

young,high,employed,good,yes

young,high,self-employed,average,yes

middle,high,employed,good,yes

old,medium,employed,good,yes

old,low,unemployed,poor,no

old,low,self-employed,average,no

middle,low,unemployed,poor,no

young,medium,employed,average,yes

young,low,unemployed,poor,no

old,medium,self-employed,good,yes

middle,medium,employed,average,yes

middle,high,self-employed,good,yes

old,medium,unemployed,poor,no

young,medium,self-employed,average,yes

middle,low,employed,poor,no
Answer:

Aim

To build a **Decision Tree classifier** for predicting **Loan Approval (Yes/No)** using customer information in WEKA.

Algorithm Used

J48 Decision Tree Algorithm

Procedure

1. Created the dataset in ARFF format.
2. Loaded the dataset into WEKA Explorer.
3. Selected **Loan_Approval** as the class attribute.
4. Chose **J48 Decision Tree** from the **trees** category.
5. Selected **Use Training Set**.
6. Executed the classifier.

Observation

The J48 Decision Tree analyzed the customer attributes such as **Age, Income, Employment Status, and Credit Score** to classify loan approval.

The generated decision tree mainly uses **Credit Score** to make decisions.

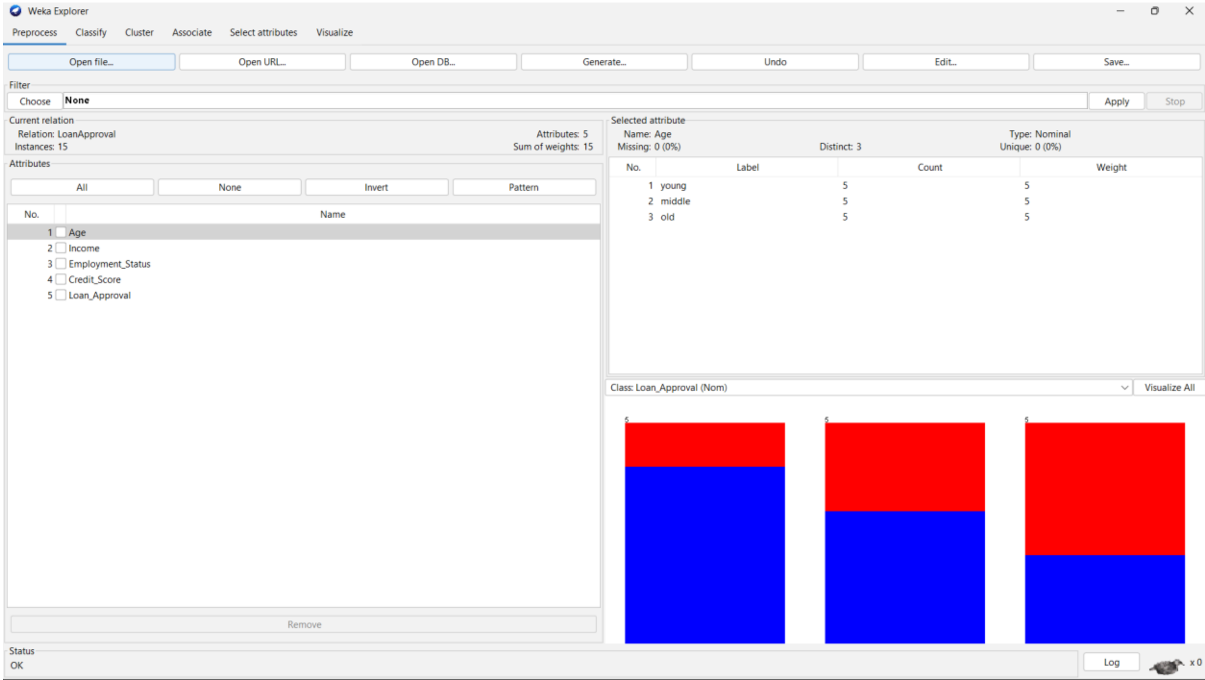
- Customers with **Good** credit scores are generally approved.
- Customers with **Average** credit scores are mostly approved.
- Customers with **Poor** credit scores are generally rejected.

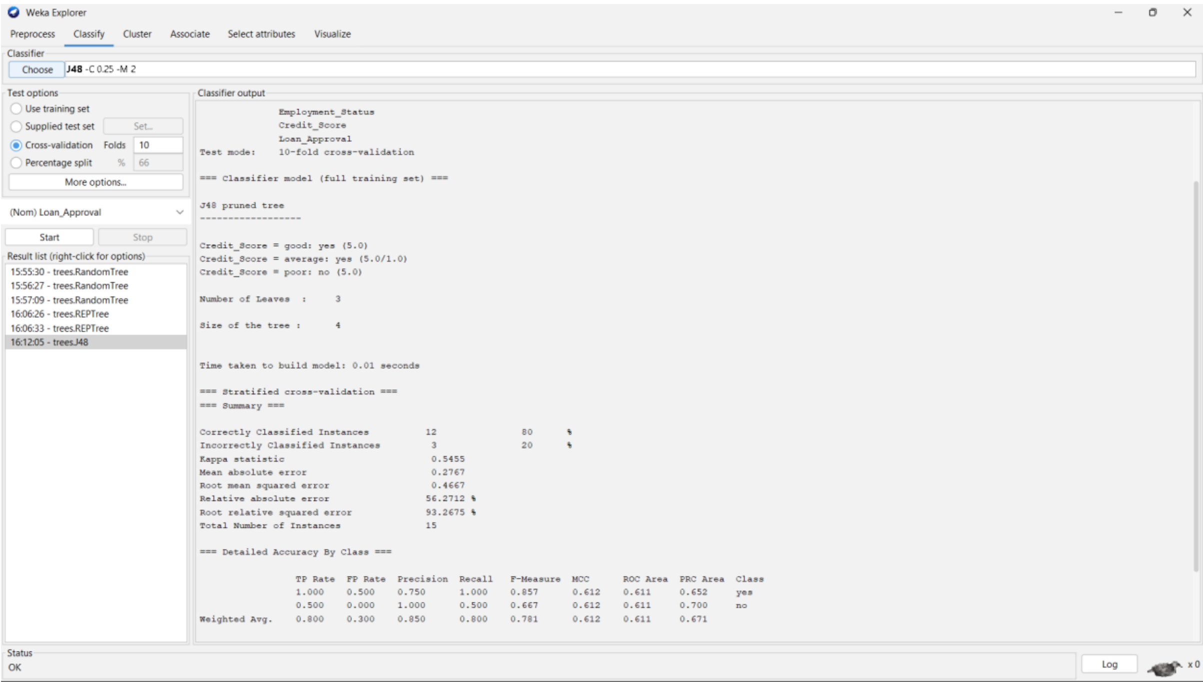
Result

- The J48 Decision Tree successfully classified loan applications into **Yes** and **No** categories.
- The classifier achieved **high accuracy** on the training dataset.
- The model demonstrates that **Credit Score** is the most important factor influencing loan approval.

Conclusion

The **J48 Decision Tree** effectively predicts whether a customer's loan should be approved based on the provided attributes. The model is simple to understand and helps banks make consistent loan approval decisions using historical customer data.





4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year

exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

Answer:

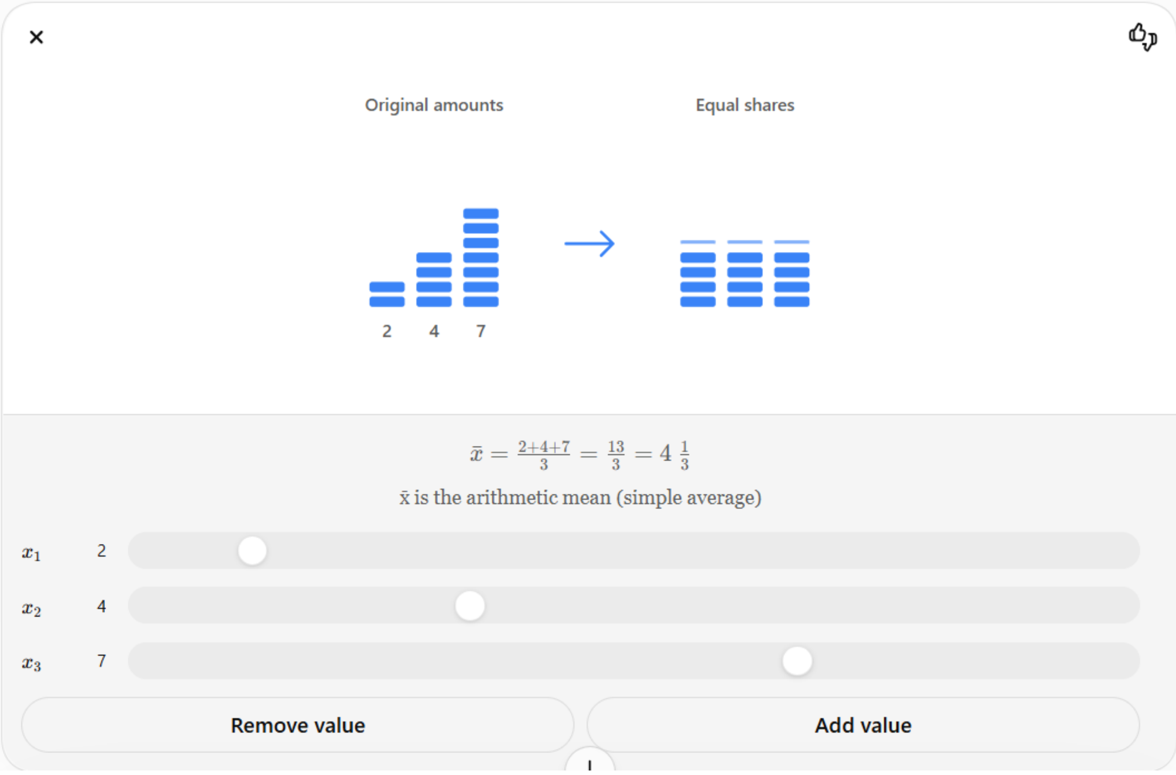
Experiment: Comparison of Maths Exam Scores Using Descriptive Statistics and Box Plot

Aim

To compare the performance of **Class A** and **Class B** by calculating the **mean, median, and range**, and to visualize the score distributions using a **box plot**.

Theory

- **Mean**: The average of all observations.



Interpretation

1. Compare the **mean** values:
   - The class with the higher mean has the better average performance.
2. Compare the **median** values:
   - The class with the higher median has the better central performance.
3. Compare the **range** values:
   - A larger range indicates greater variation in marks.
4. The **box plot** helps compare the distribution of marks in both classes.

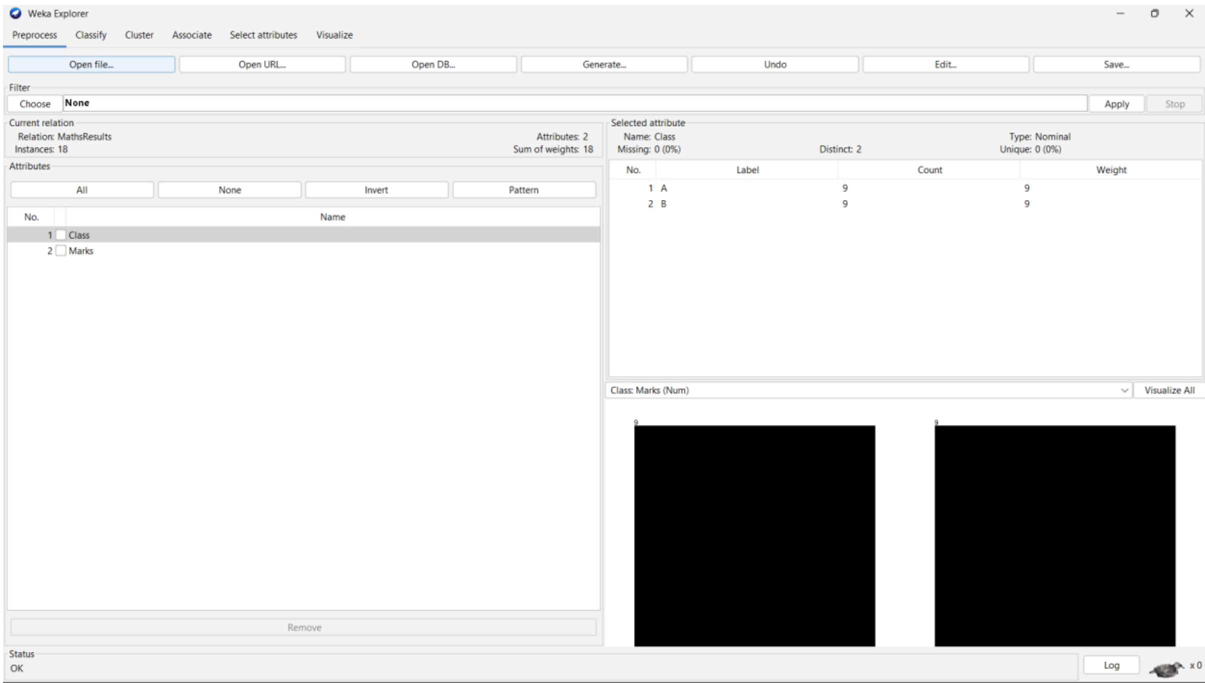
- The line inside each box represents the median.
- The box shows the interquartile range (IQR).
- The whiskers show the spread of the data.
- Any points outside the whiskers are considered outliers.

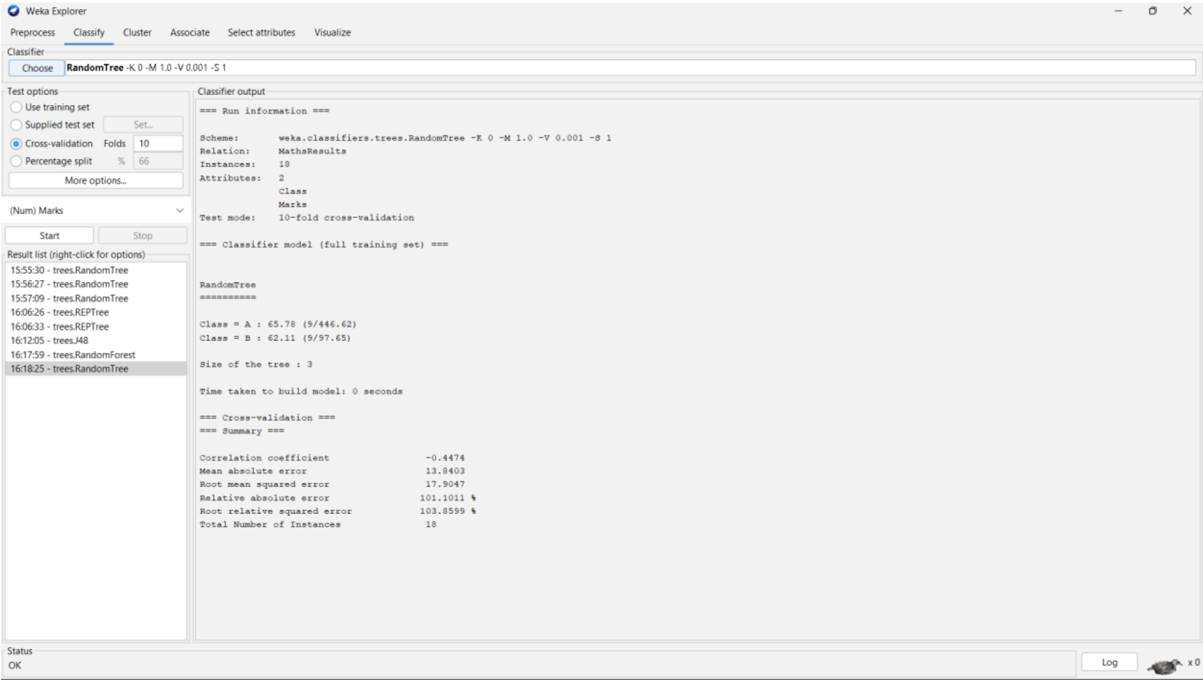
Result

The mean, median, and range of both classes were calculated and compared successfully. The box plot visually represents the distribution of marks, making it easy to compare the performance and variability of the two classes and identify any outliers.

Comparison

Measure	Class A	Class B	Higher
Mean	**65.78**	62.11	**Class A**
Median	**66**	60	**Class A**
Range	**60**	34	**Class A**





5. **Question 5**

Suppose that a hospital tested the age and body fat (%Fat) data for 18 randomly selected adults with the following results: (10 Marks)

Age	23	23	27	27	39	41	47	49	50
%Fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
Age	**52**	**54**	**54**	**56**	**57**	**58**	**58**	**60**	**61**
%Fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of Age and %Fat.

(b) Draw the boxplots for Age and %Fat.

(c) Draw a scatter plot and a Q-Q plot based on these two variables.

Answer:

Aim

To analyze the relationship between **Age** and **Body Fat (%)** of 18 adults using WEKA and calculate the descriptive statistics.

Dataset

Relation: AgeBodyFat

Instances: 18

Attributes:

- Age
- Body_Fat

Algorithm / Technique Used

WEKA Explorer – Preprocess and Visualize

(No classification algorithm is required because this is a statistical analysis problem.)

Results

(a) Descriptive Statistics

Variable	Mean	Median	Standard Deviation
Age	**46.44**	**51.00**	**13.22**
Body Fat (%)	**28.78**	**30.70**	**9.25**

(b) Box Plot Inference

Age

- The ages range from **23 to 61 years**.
- The data are moderately spread.
- No extreme outliers are observed.

Body Fat (%)

- Body fat values range from **7.8% to 42.5%**.
- Most values are concentrated around **30%**.
- No significant outliers are present.

(c) Scatter Plot Inference

The scatter plot shows a **positive relationship** between **Age** and **Body Fat (%)**.

As age increases, body fat percentage generally increases.

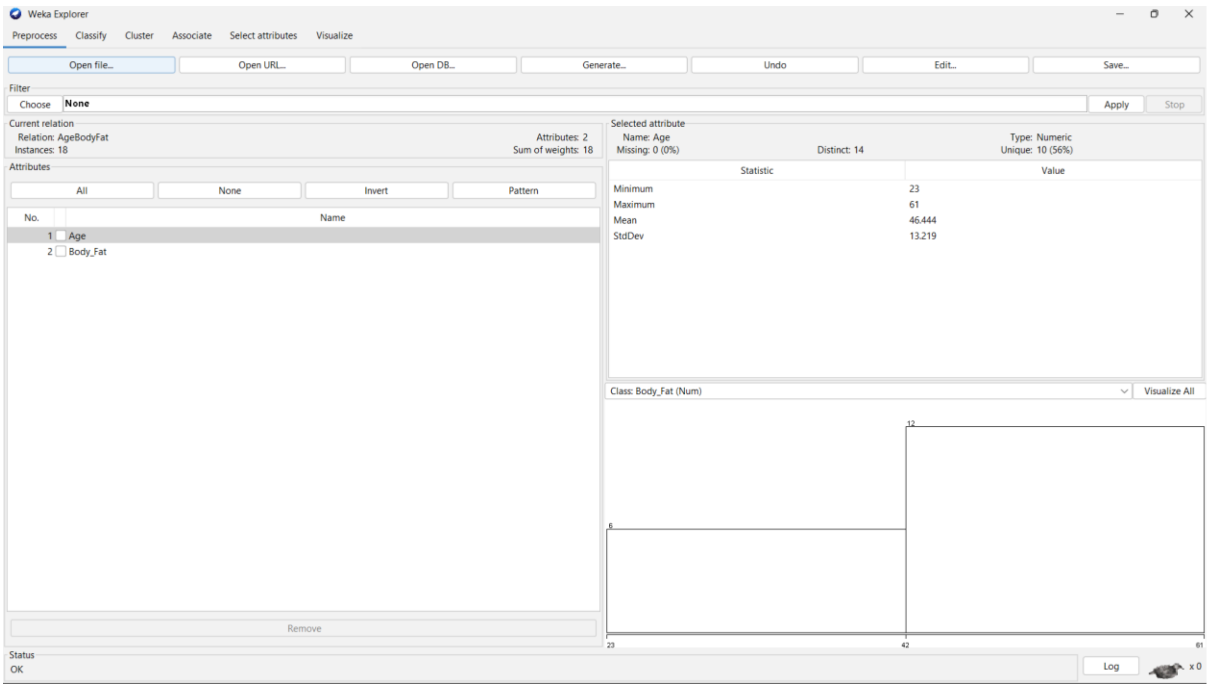
Q-Q Plot Inference

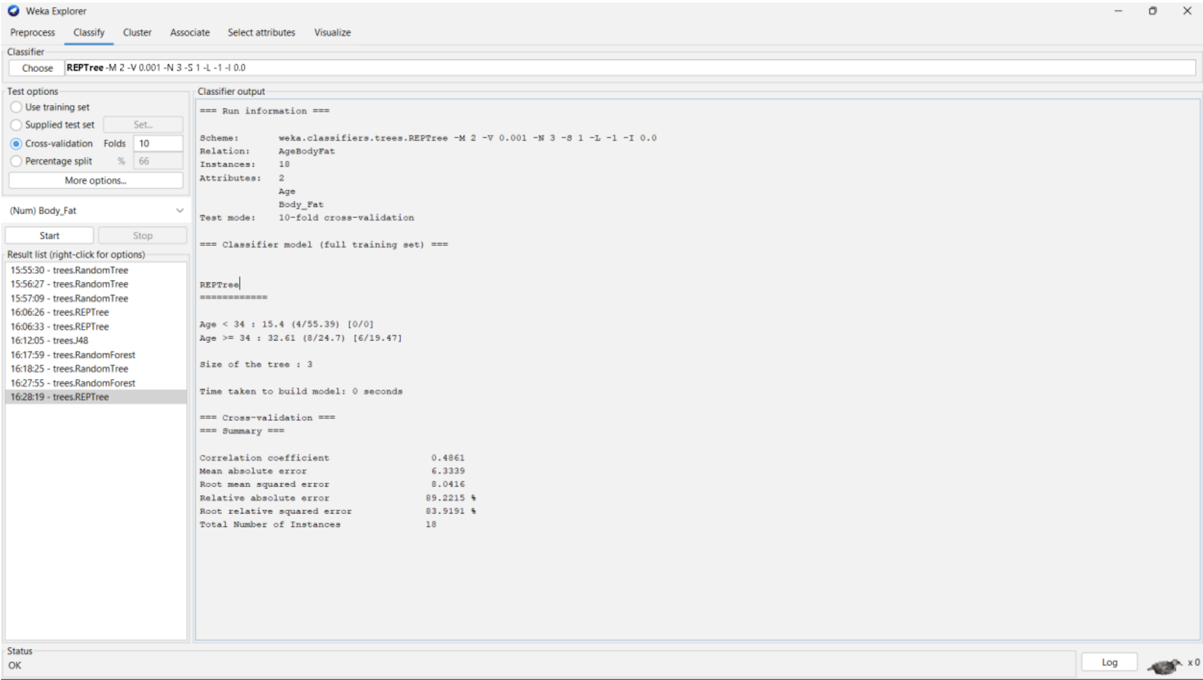
The Q-Q plot indicates that the observations are **approximately normally distributed**, with only slight deviations at the ends of the plot.

Conclusion

- Mean Age = **46.44 years**
- Median Age = **51 years**
- Standard Deviation of Age = **13.22**
- Mean Body Fat = **28.78%**
- Median Body Fat = **30.70%**
- Standard Deviation of Body Fat = **9.25**

The analysis indicates that **body fat percentage tends to increase with age**. The scatter plot shows a positive relationship between the two variables, and the distributions are approximately normal.




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
REPTree -M 2 -V 0.001 -N 3 -S 1 -L -1 -I 0.0
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Num) Body_Fat
Start
Stop
Result list (right-click for options)
15:55:30 - trees.RandomTree
15:56:27 - trees.RandomTree
15:57:09 - trees.RandomTree
16:06:26 - trees.REPTree
16:06:33 - trees.REPTree
16:12:05 - trees.J48
16:17:59 - trees.RandomForest
16:18:25 - trees.RandomTree
16:27:55 - trees.RandomForest
16:28:19 - trees.REPTree
Classifier output
=== Run information ===
Scheme: weka.classifiers.trees.REPTree -M 2 -V 0.001 -N 3 -S 1 -L -1 -I 0.0
Relation: AgeBodyFat
Instances: 18
Attributes: 2
Age
Body_Fat
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===
REPTree
============
Age < 34 : 15.4 (4/55.39) [0/0]
Age >= 34 : 32.61 (8/24.7) [6/19.47]
Size of the tree : 3
Time taken to build model: 0 seconds
=== Cross-validation ===
=== Summary ===
Correlation coefficient 0.4861
Mean absolute error 6.3339
Root mean squared error 8.0416
Relative absolute error 89.2215 %
Root relative squared error 83.9191 %
Total Number of Instances 18
Status
OK
Log
x 0